(Т3)

$$p(x) = \begin{cases} \frac{e^{-\frac{x}{\theta}}}{\theta}, & x \geq 0, \theta > 0 \\ 0, & x < 0, \theta > 0 \end{cases}$$

Выборка объема $n = 3$ $\quad \theta: \tilde{\theta}_1 = \bar{x}$; $\tilde{\theta}_3 = x_{(2)}$

I) Рассмотр $\tilde{\theta}_1$

а) Несмещенность

$$M\tilde{\theta}_1 = M\bar{x} = \frac{1}{n}\sum_{i=1}^{n} M x_i = M\xi = \theta \Rightarrow \tilde{\theta}_1 \text{ - несмещен.}$$

б) $$D\tilde{\theta}_1 = \frac{1}{n^2} n D\xi = \frac{\theta^2}{n} \overset{n=3}{\Rightarrow} D\tilde{\theta}_1 = \frac{\theta^2}{3}$$

II) Рассмотр. $\tilde{\theta}_2$

а) Несмещенность

$$f(x_{(2)}) = \frac{6}{\theta} e^{-\frac{x}{\theta}} \left(\cancel{1} - \left(\cancel{1} - e^{-\frac{x}{\theta}}\right)\right)\left(1 - e^{-\frac{x}{\theta}}\right) = \frac{6}{\theta} e^{-\frac{2x}{\theta}}\left(1 - e^{-\frac{x}{\theta}}\right)$$

$$M\tilde{\theta}_3 = \int_0^\infty x \frac{6}{\theta} e^{-\frac{2x}{\theta}}\left(1 - e^{-\frac{x}{\theta}}\right) dx = \frac{6}{\theta}\left[\int_0^\infty -\frac{\theta}{2} x \, de^{-\frac{2x}{\theta}} - \int_0^\infty x e^{-\frac{3x}{\theta}} dx\right]$$

$$= \frac{6}{\theta}\left[+\frac{\theta}{2}\int_0^\infty e^{-\frac{2x}{\theta}} dx + \frac{\theta}{3}\int -e^{-\frac{3x}{\theta}} dx\right] =$$

$$= 3\left(-\frac{\theta}{2}\right) e^{-\frac{2x}{\theta}}\Big|_0^\infty + 2\frac{\theta}{3} e^{-\frac{3x}{\theta}}\Big|_0^\infty =$$

$$= -\frac{3}{2}\theta + \frac{2}{3}\theta = \frac{5}{6}\theta \text{ - смещ}$$

Исправленная

$$\tilde{\theta}_3' = \frac{6}{5}\tilde{\theta}_3 \text{ - несмещ.}$$

б) $$M\tilde{\theta}_3^2 = \frac{6}{\theta}\int_0^\infty x^2 e^{-\frac{2x}{\theta}}\left(1 - e^{-\frac{x}{\theta}}\right) dx = \frac{6}{\theta}\int_0^\infty -\frac{\theta}{2} x^2 de^{-\frac{2x}{\theta}} + \int_0^\infty \frac{\theta}{3} x^2 de^{-\frac{3x}{\theta}} =$$

$$= \frac{6}{\cancel{\theta}}\left[+\frac{\cancel{\theta}}{\cancel{2}}\cancel{2}\left(+\frac{\theta^2}{4}\right) + \frac{\cancel{\theta}}{3} 2 \frac{\theta^2}{9}\right] = +\frac{3}{2}\theta^2 + \frac{4}{9}\theta^2 = \frac{19}{18}\theta^2$$

$$D\tilde{\theta}_3 = M\tilde{\theta}_3^2 - (M\tilde{\theta}_3)^2 = \frac{13}{36}\tilde{\theta}_3$$

$$D\tilde{\theta}_3' = \frac{36}{25} D\tilde{\theta}_3 = \frac{13}{25}\theta^2$$

<u>Эфф. оценка</u>

$\tilde{\theta}_1 \qquad \tilde{\theta}_3'$

$$\frac{\theta^2}{3} < \frac{13}{25}\theta^2 \Rightarrow$$ <u>$\tilde{\theta}_1$ более эффективная</u>

б) <u>Исследов с помощью НКР</u>

- $p(x)$ - непрер дифф при $\theta > 0$
- $$\frac{\partial}{\partial\theta}\int_0^\infty \frac{1}{\theta} e^{-\frac{x}{\theta}} dx = + \frac{\partial}{\partial\theta} 1 = 0$$

$$\int_0^\infty \frac{\partial}{\partial\theta}\left(\frac{1}{\theta} e^{-\frac{x}{\theta}}\right) dx = \int_0^\infty \left(-\frac{1}{\theta^2} e^{-\frac{x}{\theta}} + \cancel{x \frac{1}{\theta^3} e^{-\frac{x}{\theta}}}\right) dx =$$

$$= +\frac{1}{\theta} e^{-\frac{x}{\theta}}\Big|_0^\infty - \frac{1}{\theta^2}\left[\cancel{x e^{-\frac{x}{\theta}}\Big|_0^\infty} - \int_0^\infty e^{-\frac{x}{\theta}} dx\right] =$$

$$= -\frac{1}{\theta} - \frac{1}{\theta} e^{-\frac{x}{\theta}}\Big|_0^\infty = 0$$ - рав. Ф-ла соблюдается

- $$I(\theta) = M\left[\left(\frac{\partial \ln p(x)}{\partial\theta}\right)^2\right] = \int_0^\infty \left(\frac{\partial \ln p(x)}{\partial\theta}\right)^2 p(x)\, dx =$$

$$= \int_0^\infty \left(-\frac{1}{\theta} + \frac{x}{\theta^2}\right)^2 \frac{1}{\theta} e^{-\frac{x}{\theta}} dx =$$

$$= \frac{1}{\theta}\int_0^\infty \left(\frac{x^2}{\theta^4} - \frac{2x}{\theta^3} + \frac{1}{\theta^2}\right) e^{-\frac{x}{\theta}} dx = \cancel{\frac{2}{\theta^2}} - \cancel{\frac{2}{\theta^2}} + \frac{1}{\theta^2} = \frac{1}{\theta^2}$$ непрер на Θ и > 0

$\Rightarrow$ <u>Вер. мод регулярна</u>

- $\tilde{\theta}_1$ - несмещ, дисперсия ограничен. на $\forall$ компакте из Θ модель регулярна $\Rightarrow$ $\tilde{\theta}_1$ яв-ся <u>регул. оценкой</u>
- $\tilde{\theta}_3'$ - несмещ, дисп. огран на $\forall$ компакте из Θ, мод регул $\Rightarrow$ $\Rightarrow$ <u>регул оценка</u>

- Нер во Крамера - Рао

- $\tilde{\theta}_1$

$D\tilde{\theta}_1 = \frac{\theta^2}{3} \geq \frac{1}{3\,\frac{1}{\theta^2}} = \frac{\theta^2}{3}$ $\forall \theta \in \Theta$ => по дост усл эф

=> $\tilde{\theta}_1$ - эффективн оценкой

- $D\tilde{\theta}_3 = \frac{13}{25}\theta^2 \neq \frac{\theta^2}{3}$ => по ① единств $\tilde{\theta}_3$ - не яв-ся эффективн оценкой